communications engineering

A Nature Portfolio journal

Comment

https://doi.org/10.1038/s44172-025-00430-5

Orbital debris requires prevention and mitigation across the satellite life cycle

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Satellites are crowding orbital space, particularly Low Earth Orbit, exacerbating the risk of orbital debris. Here, measures across the satellite life cycle are recommended to promote space sustainability including stronger regulations, fiscal and market-based interventions, multilateral institutions, and active debris removal.

Commercialization of space, falling launch costs, satellite miniaturization, and the creation of megaconstellations are quickly increasing the number of orbiting spacecraft, particularly in Low Earth Orbit (LEO)[1]. As usage of the finite resource of orbital space grows, so does the risk of orbital debris, defined by the Inter-Agency Space Debris Coordination Committee as "all man-made objects including fragments and elements thereof, in Earth orbit or re-entering the atmosphere, that are nonfunctional"[2]. Such objects range from decades-old defunct satellites[3] to trash bags, rocket bodies[4–6], and thousands of pieces produced by accidental collisions and anti-satellite tests[7]. 45,000 objects over 10 cm^3 in size are actively tracked by the U.S. Department of Defense's Space Surveillance Network[8], while over 130 million objects ranging from 1 mm to 1 cm remain untracked[9]. All of this fast-moving debris threatens spacecraft operations and space sustainability. It also raises fears of the Kessler syndrome, or the potential for cascading collisions to trigger a positive feedback loop that creates a debris field rendering certain orbits unusable[10] and impeding human space activities[11]. This scenario could be reached in 50–100 years without active debris removal and mitigation, and in less than 200 years even with such interventions[12]. Collisions are expected to become the primary process generating debris, particularly 900–1000 km above Earth, which has a high density of satellites[5].

Promisingly, two key space agencies have adopted orbital debris mitigation measures[13]. The U.S. Federal Communications Commission (FCC), which regulates U.S. satellites and their communications, and the European Space Agency (ESA) each now mandate that LEO satellites deorbit within five years of mission completion[14,15]. Still, orbital debris is proliferating, and global adherence to the voluntary 25-year disposal norm remains lax[16] with some exceptions[17]. To reduce orbital debris, prevention and mitigation measures across the satellite life cycle from pre-launch to launch, operations, and retirement are crucial. These include stronger regulatory policies and fiscal and market-based mechanisms, new and expanded multilateral institutions, and investment in technologies like active debris removal.

Limiting the generation of new orbital debris pre-launch

Satellite licensing agencies should prioritize approvals based on a framework integrating social, environmental, and economic benefits and risks, as was mooted in 2000 during discussions at the United Nations Committee on the Peaceful Uses of Outer Space[18]. Such a framework would resemble the World Bank's consideration of methods to prioritize infrastructure investment[19]. A satellite prioritization process might rank a fire detection satellite above a commercial television satellite or deprioritize a massive satellite in LEO due to higher associated collision risks. The NASA Space Technology Mission Directorate's civil space challenges ranking exercise could serve as a model for developing a prioritization framework with input from civil, commercial, academic, and other stakeholders. Cross-sector collaboration would help build only the most necessary satellites and reduce speculative bubbles, at least within single markets[20]. This could help avoid potentially environmentally detrimental or wasteful situations, such as the bankruptcies of satellite phone companies Iridium and Globalstar in 1999 and 2000, respectively[21]. Iridium was ordered by a court to destroy its 66 satellites, a situation that was only avoided due to an effective bailout by the U.S. Department of Defense, which began contracting with the bankrupt company for its services in 2000. Yet as geopolitical tensions heighten, more countries may seek to ensure the resilience of their critical infrastructure by operating their own satellites. China and the European Union, for instance, are each working to develop an alternative to Starlink, the LEO satellite internet megaconstellation owned by US aerospace company SpaceX. Such efforts could result in redundant megaconstellations, which, while potentially bolstering national security, could decrease space sustainability by raising the risks of orbital debris.

Single markets such as the European Union or the U.S. should also test fiscal and financial solutions to orbital debris, which represents a negative externality since its costs are not borne solely by the actor who generates it. Governments should implement orbital-use fees such as those suggested by Rao et al.[22]. These would incentivize operators to use limited orbits more efficiently and factor in collision risk while promising, if internationally harmonized, to quadruple the value of the global satellite industry by 2040. Governments should also consider imposing taxes on satellite operators at launch or issuing penalties based on the amount of debris generated[23]. The FCC's first space debris enforcement action in 2023, when it fined a satellite operator, DISH, for improper satellite decommissioning[15], exemplifies such a penalty.

Launch companies should also be taxed, especially as their operations grow more frequent. A launch tax proposed by the former Biden administration[24] should be implemented, with revenues going to fund a new U.S.-led space traffic management organization (STMO) or debris remediation technologies. Regulatory agencies should also require operators to obtain orbital debris insurance. This would build on policy in the U.S., where the Federal Aviation Administration, which regulates commercial space transportation, requires launch licensees to have insurance covering third-party and government property damage claims. Premiums from orbital debris insurance could be collected until any upper-stage rockets or satellites are passivated[25]. A portion of the premiums collected by insurers could underwrite a fund that finances orbital safety and sustainability, drawing inspiration from the World Bank's International Finance Corporation's

partnership with 14 global insurance companies to boost infrastructure investment in emerging markets[26]. One additional market-based solution that governments should encourage is tradeable satellite performance bonds, the values of which could reflect the sustainability of a satellite's operations[27] or the success of its deorbiting[28].

To encourage conscientious use of finite terrestrial and orbital resources, governments should promote a circular space economy, which embraces waste reduction, reuse, and recycling, through policy and technology innovations. In the U.S., reusability has been embraced by commercial aerospace companies like SpaceX, whose reusable rockets have helped to minimize the environmental footprint of manufacturing components[29]. Yet by lowering the costs and increasing the frequency of launch, reusable rockets enable spacecraft to be slotted into orbit at a potentially unsustainable rate. To avoid excessive single-use missions, government support for research into on-orbit servicing, assembly, and manufacturing is key[30].

Compared to the U.S., the ESA has initiated more top-down measures to build a circular economy in space by 2050. In 2012, it launched its Clean Space initiative to develop technologies for deorbiting systems, design for demise, and passivation. One ESA-commissioned program, Clearspace-1, aims to remove a 112-kg rocket part from orbit in 2028[31]. These efforts may stimulate a European market for in-orbit servicing and debris removal and foster a circular economy for orbital debris, whose lessons could hold for Earth[32,33] and other celestial bodies. Already, orbital debris is accumulating around the moon, where dozens of near-collisions have been reported[34].

Improving orbital safety during launch and operations

STMOs are crucial to managing space traffic in the global commons of orbital space[35]. Encouragingly, their creation appears to be underway in Europe as indicated by a 2023 RAND report[31]. Within an STMO, as Reilly et al.[36] suggest, members should share data on space situational awareness, space vehicle status, and space weather, establish shared norms and regulations around liability, collision avoidance, and passivation of satellites and upper stages, and cooperate to make space traffic safe and sustainable[36]. It is also recommended that STMOs involve commercial actors, given their growing role in space traffic, especially in the U.S., where SpaceX launches the majority of payloads. STMOs are necessary not only to make satellite operations safer, but because the growing frequency of launches and returning objects threatens to interfere with air traffic, too[37]. Space traffic occurs at much higher speeds than air traffic and is generally unmanned, making space situational awareness and well-coordinated, automated systems for collision avoidance all the more important. While Jakhu et al. suggest expanding the mandate of the International Civil Aviation Organization (ICAO), a UN body established in 1947, beyond civil aviation to include space traffic[38], given the fracturing international order, STMOs are likelier to first be established by groups of spacefaring nations such as ESA members or signatories of the Artemis Accords, which were drafted by NASA and the U.S. Department of State and which have been signed by 54 countries as of April 2025. STMO member contributions could be modeled on a formula reflecting the combined mass of their civil and commercial spacecraft, similar to how International Maritime Organisation (IMO) member state contributions reflect their merchant fleet's tonnage.

New STMOs should also strive to develop automated collision avoidance systems. The ESA performs at least one collision avoidance maneuver per satellite per year, largely to steer clear of orbital debris, by manually uploading commands to a satellite when its chance of collision is estimated to exceed 1/10,000. The ESA's Space Safety Program is also working with the private sector to develop technologies for collision avoidance. Starlink satellites, which have onboard automated collision avoidance systems fed by U.S. Space Surveillance Network data, make far more maneuvers than ESA satellites, partly as a result of following a much higher collision avoidance threshold of 1/100,000[39]. Still, Starlink satellites nearly struck the ESA's Aeolus satellite in 2019[40] and the China Space Station twice in 2021, prompting their operators to perform evasive maneuvers. After the incidents, China began publishing daily information about its space station's orbit to benefit collective collision avoidance[41]. SpaceX, Starlink's operator, began publishing ephemerides, or data on positioning and timing, for the megaconstellation's satellites in 2021[42].

All spacecraft operators must continue to improve communication and transparency, expand sharing of space surveillance data, and reach agreements on collision risk thresholds and norms[43]. Ultimately, to be effective, an STMO should possess the authority to coordinate and oversee the navigation of space objects under its domain, whether civilian or commercial, so that two objects trying to out-maneuver one another do not accidentally collide. When a collision is anticipated, it is crucial that the STMO can decide which object should move. Ideally, this should be the lighter one to save fuel. The question of authority over spaceborne military objects, however, would likely remain fraught even with the establishment of STMOs. The risk of military objects colliding could also rise as militarization and weaponization of space proceed. This threat underscores the need for communication and transparency between spacefaring actors[44], including between civilian and military actors to the extent possible.

Mitigating extant orbital debris post-launch

To mitigate orbital debris, more robust regulations at national and international scales and new technological solutions are necessary. Strengthening national and supranational regulations is the most practical way to begin tackling orbital debris. Initially, the U.S. led the way by introducing national orbital debris mitigation guidelines in 1979 with the founding of NASA's Orbital Debris Program, which was spurred by the visible reentries and breakups of Kosmos 954, a nuclear-powered Soviet reconnaissance satellite, and Skylab, the first U.S. space station[45]. Yet these early efforts have been contradicted by the rapid and lightly regulated development of the country's commercial space sector. NASA also does not appear to view mitigating orbital debris as a priority compared to other issues[46]. This relative disregard was confirmed by the agency's cancellation of its On-orbit servicing, Assembly, and Manufacturing 1 (OSAM-1) mission, which had been investigating how to refuel satellites to extend their missions[47]. Still, U.S. regulations like the U.S. Government Orbital Debris Mitigation Standard Practices, first released in 2001 and last updated in 2019[48], show some continued leadership domestically and abroad. Internationally, the U.S. became the first country to ban testing of direct-ascent anti-satellite missile systems in 2022, albeit only after decades of conducting its own tests[49]. The Artemis Accords call upon signatories to avoid generating orbital debris by properly passivating and disposing of their spacecraft post-mission. Yet these initiatives are undermined by the FCC's approval of 42,000 Starlink satellites and insistence that commercial megaconstellations are excluded from environmental review[50], despite recommendations from the U.S. Government Accountability Office in 2022 to revisit this policy[51]. Maintaining this exemption sets a poor example for other spacefaring nations seeking to launch LEO megaconstellations.

The European Union (EU) and ESA, which represents 23 member states as of May 2025, are global leaders in space sustainability, setting higher targets than the U.S. In 2025, the EU plans to introduce its Space Law, which will likely require compliance from all companies providing services within its market, including non-EU actors. The EU Space Law is anticipated to require commercial satellite operators to follow rules across the satellite life cycle, from launch to collision avoidance, information sharing, and deorbiting. The ESA

is increasingly focusing on sustainability, with its Space Debris Office publishing an annual Space Environment Report on international orbital debris management and space sustainability since 2016[9]. As part of the agency's push for a circular economy, the ESA Space Debris Mitigation Requirements aim for a net-zero pollution strategy in space by 2030. The ESA has also suggested extending protected regions, or the areas in which the generation of space debris is tightly regulated to ensure their sustainability, beyond LEO and geostationary orbits to medium Earth orbits used by Global Navigation Satellite Systems (~20,000–23,000 km above Earth) and lunar orbits.

To popularize its norms, the ESA is promoting its Zero Debris Charter, signed by over two dozen European space agencies, research institutes, and space companies[52]. The ESA should also work with the FCC to internationalize their shared five-year post-operation deorbiting rule and make it mandatory. Bilateral efforts could start by targeting the Inter-Agency Space Debris Coordination Committee (IADC), whose creation can be traced to ESA-NASA coordination meetings in 1987[53]. The IADC remains the leading international body focused on orbital debris. Presently, the IADC Space Debris Mitigation Guidelines are weaker than the ESA and FCC rules, expecting but not requiring objects to have a residual end-of-mission orbital lifetime of <25 years, though this is shorter for large constellations (>100 satellites). Several national governments have codified the IADC guidelines, which were also the model for the UN Committee on the Peaceful Uses of Outer Space (COPUOS) 2007 Space Debris Mitigation Guidelines. Internationally, however, they remain voluntary.

For its part, the IADC, which includes eleven additional space agencies such as Japan Aerospace Exploration Agency, Roscosmos, and China National Space Agency, should expand to better represent the growing number of spacefaring actors. Commercial actors and non-spacefaring states, many of which may eventually operate spacecraft, are two obvious candidate categories. The IADC could make adherence to ISO Standard 24113: Space Debris Mitigation Requirements a requirement for companies to join[54]. Efforts are necessary to ensure inclusive and transparent ISO standard-setting, for Western Europe dominance has been shown to impede adoption of standards by small-scale producers and developing countries[55]. An expanded IADC could draw inspiration from ICAO, which invites international organizations to attend relevant meetings, and the IMO, which offers consultative status to non-governmental international organizations. Both organizations offer technical training to member states in aviation and safe shipping, respectively. The IADC should develop similar programs relevant to orbital space, which would enable knowledge exchange and technology transfer between legacy and emerging satellite operators. The IADC should also encourage member states to require environmental impact reviews for satellites, particularly those over a certain weight or constellations over a certain size, or for those occupying orbits or regions sensitive to space debris. If committed to the task of becoming a global standard-setter in orbital space, the IADC should investigate the international feasibility of a maneuverability requirement for satellites operating over 400 km above Earth, as the FCC was assessing as of 2024[56].

Even if all launches were to cease, the sheer volume of objects already in orbit means that the Kessler Syndrome would remain a concern. To reduce extant debris, although NASA has canceled OSAM-1, other spacefaring actors are exploring technological solutions. RemoveDEBRIS, an in-orbit active debris removal demonstration mission, involves partners from the United Kingdom, several European countries, and South Africa[57]. In 2022, China Aerospace Science and Technology Corporation undertook the first-ever space tug operation by docking with a defunct BeiDou navigation satellite and pulling it from geosynchronous to graveyard orbit[58]. Tokyo-based Astroscale, an orbital debris removal company, has also made headway in attempting to remove an unprepared Japanese upper-stage rocket body[59]. To support international interest in active debris removal and overcome the technology's high costs, which are difficult for any single actor to bear[60], the IADC should establish and administer a financial mechanism to bolster collaborative research and development. This mechanism could solicit contributions from countries, the private sector, and philanthropic organizations in a manner resembling the UN's Loss and Damage Fund for Developing Countries[61]. Finally, for active debris removal to become legally viable, spacefaring actors will need to agree on an explicit definition of orbital debris and the circumstances in which it can be removed. An expanded IADC would be a relevant venue for such discussions, ideally involving both government and commercial stakeholders.

Conclusion

Across the satellite life cycle, satellite prioritization, fiscal and market-based solutions, strengthened and expanded regulatory bodies, and technological innovations are key to preventing and mitigating orbital debris. Most measures are likely to be implemented at the national or supranational scale before they can be extended and harmonized around the world. With the U.S. seemingly unwilling to reign in its commercial satellite sector, the ESA stands to become a global leader in space sustainability. The IADC should expand its membership and set international norms and standards. All spacefaring actors should work to prevent and mitigate orbital debris before the Kessler Syndrome becomes a reality. Recommended interventions are:

- *Before launch*: Licensing agencies should prioritize satellite approvals according to cost-benefit analysis, encourage fiscal and market-based solutions, including taxes, fees, and bonds, and promote reusability and a circular space economy
- *During launch and operations*: STMOs should be formed and collision avoidance technologies and standards developed and shared to make launch, operations, and return safer for both space and air traffic
- *After launch*: Stricter national regulations and international harmonization through organizations like an expanded IADC and technological innovations in active debris removal should be promoted to reduce extant orbital debris.

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Received: 24 January 2024; Accepted: 12 May 2025;
Published online: 26 May 2025

References

1. Pardini, C. & Anselmo, L. Environmental sustainability of large satellite constellations in low Earth orbit. *Acta Astronaut.* **170**, 27–36 (2020).
2. Inter-Agency Space Debris Coordination Committee. *IADC Space Debris Mitigation Guidelines* https://www.unoosa.org/res/oosadoc/data/documents/2025/aac_105c_12025crp/aac_105c_12025crp_9_0_html/AC105_C1_2025_CRP09E.pdf (2020).
3. Hall, L. The history of space debris. In *Space Traffic Management Conference* Vol. 19 https://commons.erau.edu/stm/2014/thursday/19/ (2014).
4. NASA Orbital Debris Program Office. *History of On-Orbit Satellite Fragmentations* 16th edn https://ntrs.nasa.gov/api/citations/20220019160/downloads/HOOSF_16e_all_for_STRIVES.pdf (2022).
5. Liou, J. C. & Johnson, N. L. Risks in space from orbiting debris. *Science* **311**, 340–341 (2006).
6. Anselmo, L. & Pardini, C. Ranking upper stages in low Earth orbit for active removal. *Acta Astronaut.* **122**, 19–27 (2016).
7. Pardini, C. & Anselmo, L. Physical properties and long-term evolution of the debris clouds produced by two catastrophic collisions in Earth orbit. *Adv. Space Res.* **48**, 557–569 (2011).
8. NASA. *State-of-the-Art of Small Spacecraft Technology* https://www.nasa.gov/wp-content/uploads/2025/02/soa-2024.pdf (2024).
9. European Space Agency. Space environment statistics https://sdup.esoc.esa.int/discosweb/statistics/ (2024).

10. Kessler, D. J. & Cour-Palais, B. G. Collision frequency of artificial satellites: the creation of a debris belt. *J. Geophys. Res.* **83**, 2637–2646 (1978).
11. Migaud, M. R. Protecting Earth's orbital environment: policy tools for combating space debris. *Space Policy* **52**, 101361 (2020).
12. Nomura, K. et al. Tipping points of space debris in low Earth orbit. *Int. J. Commons* **18**, 17–31 (2024).
13. European Space Agency Debris Office. *ESA's Annual Space Environment Report* https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf (2025).
14. Masson-Zwaan, T. & Johnson, C. D. Echostar-7: the US imposes first-ever fine for failure to comply with deorbiting plan. *Air Space Law* **49**, 243–248 (2024).
15. European Space Debris Mitigation Working Group. *ESA Space Debris Mitigation Requirements* https://technology.esa.int/upload/media/ESA-Space-Debris-Mitigation-Requirements-ESSB-ST-U-007-Issue1.pdf (2023).
16. Adilov, N., Alexander, P. & Cunningham, B. Understanding the economics of orbital pollution through the lens of terrestrial climate change. *Space Policy* **59**, 101471 (2022).
17. NASA Office of Inspector General. *NASA's Efforts to Mitigate the Risks Posed by Orbital Debris.* Report No. IG-21-011 (NASA Office of Inspector General, Washington, DC, 2011).
18. United Nations Committee on the Peaceful Uses of Outer Space. *Legal Subcommittee, 626th Meeting, Vienna, Austria* https://www.unoosa.org/pdf/reports/transcripts/legal/LEGALT_626E.pdf (2000).
19. Marcelo, D., Mandri-Perrott, C., House, S., & Schwartz, J. *Prioritizing Infrastructure Investment: A Framework for Government Decision Making*. Policy Research Working Paper **7674** (World Bank Group, Washington, DC, 2016).
20. Kim, M. J. The potential speculative bubble in the US commercial space launch industry. *N. Space* **6**, 156–183 (2018).
21. Ailor, W. Protecting the LEO environment. *J. Space Saf. Eng.* **9**, 449–454 (2022). (2022).
22. Rao, A., Burgess, M. G. & Kaffine, D. Orbital-use fees could more than quadruple the value of the space industry. *Proc. Natl Acad. Sci. USA* **117**, 12756–12762 (2020).
23. Bernhard, P., Deschamps, M. & Zaccour, G. Large satellite constellations and space debris. *Eur. J. Oper. Res.* **304**, 1140–1157 (2023).
24. Kim, M. Biden takes aim at SpaceX's tax-free ride in American airspace. *The New York Times* https://www.nytimes.com/2024/04/04/us/politics/spacex-biden-musk-taxes.html (2024, April 4).
25. Pelton, J. N. *New Solutions for the Space Debris Problem* (Springer International Publishing, Cham, 2015).
26. International Finance Corporation. IFC mobilizes $3 billion from leading global insurers to boost real sector investments in emerging markets. https://www.ifc.org/en/pressroom/2024/ifc-mobilizes-3-billion-from-leading-global-insurers-to-boost-real-sector-investments-in-emerging-markets (2024).
27. McClintock, B. et al. *The Time for International Space Traffic Management Is Now* (RAND Corporation, Santa Monica, 2023).
28. Adilov, N., Alexander, P. J. & Cunningham, B. M. The economics of satellite deorbiting performance bonds. *Econ. Lett.* **228**, 111150 (2023).
29. Miraux, L., Wilson, A. R. & Calabuig, G. J. D. Environmental sustainability of future proposed space activities. *Acta Astronaut.* **200**, 329–346 (2022).
30. Leonard, R. & Williams, I. D. Viability of a circular economy for space debris. *Waste Manag.* **155**, 19–28 (2023).
31. European Space Agency. ClearSpace-1. https://www.esa.int/Space_Safety/ClearSpace-1 (2025).
32. Paladini, S., Saha, K. & Pierron, X. Sustainable space for a sustainable Earth? Circular economy insights from the space sector. *J. Environ. Manag.* **289**, 112511 (2021).
33. Lawrence, A. et al. The case for space environmentalism. *Nat. Astron.* **6**, 428–435 (2022).
34. Barakat, B. F. & Kezirian, M. T. Establishing requirements for lunar and cislunar orbital debris tracking. *J. Space Saf. Eng.* **11**, 446–453 (2024).
35. von der Dunk, F. G. Space traffic management: a challenge of cosmic proportions. *Proc. Int. Inst. Space Law 2015* **58**, 385–396 (2016).
36. Reilly, J., Blackford, M., & Mueller, J. Challenges in space traffic management. In *Proc. Advanced Maui Optical and Space Surveillance Technologies Conference* 1–12 https://amostech.com/TechnicalPapers/2023/Poster/Reilly.pdf (2023).
37. Georgiadis, P., Hollinger, P. & Bott, I. Rockets, drones and flying taxis: Brace for a 'Wild West' in the skies. *The Financial Times* (16 February 2025).
38. Jakhu, R. S., Sgobba, T., & Dempsey, P. S. (Eds.). *The Need for an Integrated Regulatory Regime for Aviation and Space: ICAO for Space?* (Springer, Vienna, Springer-Verlag, 2011).
39. Boley, A. & Byers, M. Anti-satellite weapon tests to disrupt large satellite constellations. *Nat. Astron.* **8**, 10–12 (2024).
40. Alfano, S., Oltrogge, D., Krag, H., Merz, K. & Hall, R. Risk assessment of recent high-interest conjunctions. *Acta Astronaut.* **184**, 241–250 (2021).
41. Lan, C. The Starlink-China Space Station near-collision: questions, solutions, and an opportunity. *The Space Review* (2022, February 28).
42. Liu, A., Xu, X., Xiong, Y. & Yu, S. Maneuver strategies of Starlink satellite based on SpaceX-released ephemeris. *Adv. Space Res.* **74**, 3157–3169 (2024).
43. Spencer, D. B., Sorge, M. E. & Skinner, M. A. Establishing "norms of behavior" for satellite collision avoidance maneuver planning. *J. Space Saf. Eng.* **11**, 120–126 (2024).
44. Hammack, K. International relations in space: the role of miscalculation, militarization, and weaponization. *Astropolitics* **19**, 230–236 (2021).
45. Portree, D. S. & Loftus, Jr., J. P. *Orbital Debris: A Chronology*. NASA/TP-1999-208856 (NASA, Washington, DC, 1999).
46. NASA Space Technology Mission Directorate. *Stakeholder Webinar: Civil Space Shortfall Ranking*. https://www.nasa.gov/wp-content/uploads/2024/07/shortfall-ranking-results-july-2024-508-tagged.pdf?emrc=fd191e (2024).
47. NASA. On-orbit Servicing, Assembly, and Manufacturing 1 (OSAM-1) https://www.nasa.gov/mission/on-orbit-servicing-assembly-and-manufacturing-1/ (2024).
48. NASA. Orbital Debris Mitigation Standard Practices, November 2019 Update https://orbitaldebris.jsc.nasa.gov/library/usg_orbital_debris_mitigation_standard_practices_november_2019.pdf. (2019).
49. Weeden, B. U.S. direct ascent anti-satellite testing. (Secure World Foundation, 2023). https://swfound.org/media/207610/fs23-07_us-da-asat-testing_0723.pdf.
50. Ryan, R. J. The fault in our stars: Challenging the FCC's treatment of commercial satellites as categorically excluded from review under the National Environmental Policy Act. *Vanderbilt J. Entertain. Technol. Law* **22**, 923–950 (2019).
51. U.S. Government Accountability Office. *Satellite Licensing: FCC Should Reexamine Its Environmental Review Process for Large Constellations of Satellites* https://www.gao.gov/assets/gao-23-105005.pdf (2022).
52. Hornyak, T. Cleaning up the final frontier. *Res. Technol. Manag.* **67**, 8–10 (2024).
53. Johnson, N. Origin of the Inter-Agency Space Debris Coordination Committee. https://ntrs.nasa.gov/api/citations/20150003818/downloads/20150003818.pdf (2015).
54. Kato, A. et al. Standardization by ISO to ensure the sustainability of space activities. In *Proc. Sixth European Conference on Space Debris, Darmstadt, Germany, 22-25 April 2013* (European Space Operations Centre, Darmstadt, 2013).
55. Zhao, X., Castka, P. & Searcy, C. ISO standards: a platform for achieving sustainable development goal 2. *Sustainability* **12**, 9332 (2020).
56. Federal Communications Commission. 47 CFR Parts 5, 25, and 97. Space Innovation; Mitigation of Orbital Debris in the New Space Age. https://www.federalregister.gov/documents/2024/08/09/2024-17093/space-innovation-mitigation-of-orbital-debris-in-the-new-space-age (2024).
57. Forshaw, J. L. et al. RemoveDEBRIS: an in-orbit active debris removal demonstration mission. *Acta Astronaut.* **127**, 448–463 (2016).
58. Ali, S. M. *The US-China-Russia Triangle: An Evolving Historiography* (Springer International Publishing, Cham, 2022).
59. Harvey, B. *Japan in Space: Past, Present and Future* (Springer International Publishing, Cham, 2023).
60. Emanuelli, M. et al. Conceptualizing an economically, legally, and politically viable active debris removal option. *Acta Astronaut.* **104**, 197–205 (2014).
61. Tietjen, B. & Gopalakrishnan, T. Loss and damage funding in the UN climate negotiations: from dialogue to reality. *Environ. Sci. Policy Sustain. Dev.* **65**, 18–28 (2023).

Competing interests

The author declares no competing interests. No, I declare I have no competing interests as defined by *Communications Engineering*, nor other interests that might be perceived to influence the interpretation of the commentary.

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Peer review information *Communications Engineering* thanks Nikola Schmidt, Joseph N. Pelton, and the other anonymous reviewer for their contribution to the peer review of this work.

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